

The Ramsey Growth Model

Optimization, dynamics, and growth properties

From Solow to Ramsey

- Consumption and saving are determined by optimizing households.
- Firms maximize profits and interact with households in competitive markets.
- The saving rate is endogenous rather than fixed.

We first set up the model, then derive its dynamic equations and study the optimal development path.

Production and notation

$$\begin{aligned}Y_t &= (A_t L_t)^{1-\alpha} K_t^\alpha, & 0 < \alpha < 1, \\A_t &= A_0 e^{xt}, & \dot{A}_t/A_t = x, \\L_t &= L_0 e^{nt}, & \dot{L}_t/L_t = n.\end{aligned}$$

Variables per person: $k_t = K_t/L_t$, $c_t = C_t/L_t$.

$$\hat{k}_t = \frac{K_t}{A_t L_t} = \frac{k_t}{A_t}, \quad \hat{c}_t = \frac{C_t}{A_t L_t} = \frac{c_t}{A_t}, \quad \hat{y}_t = \frac{Y_t}{A_t L_t} = \hat{k}_t^\alpha.$$

A dot denotes a time derivative; a hat denotes units per effective worker.

The firm's problem

$$\max_{K_t, L_t} \Pi_t = Y_t - w_t L_t - r_t K_t.$$

The first-order conditions imply

$$w_t = (1 - \alpha) A_t^{1-\alpha} L_t^{-\alpha} K_t^{\alpha} = (1 - \alpha) A_t \hat{k}_t^{\alpha}, \quad (3)$$

$$r_t = \alpha (A_t L_t)^{1-\alpha} K_t^{\alpha-1} = \alpha \hat{k}_t^{\alpha-1}. \quad (4)$$

r_t is the rental rate of capital; the net return is $r_t - \delta$. Constant returns to scale imply $w_t L_t + r_t K_t = Y_t$.

The household's problem

$$\max_{\{\hat{c}_t, \hat{k}_t\}_{t \geq 0}} \int_0^{\infty} \frac{\hat{c}_t^{1-\theta}}{1-\theta} e^{-[\rho-n-(1-\theta)x]t} dt.$$

Subject to

$$\hat{c}_t + \hat{i}_t = w_t/A_t + r_t \hat{k}_t, \quad (6)$$

$$\dot{\hat{i}}_t = \dot{\hat{k}}_t + (\delta + n + x) \hat{k}_t. \quad (7)$$

$\hat{k}_0$ is given. Assume $\theta > 0$, $\rho > 0$, and $\rho - n - (1 - \theta)x > 0$.

Aside follows: the next slides explain this objective and constraint.

Aside: household utility

There are H identical, infinitely lived households. Each has L_t/H members.

$$U = \int_0^{\infty} u(c_t) \frac{L_t}{H} e^{-\rho t} dt.$$

- c_t : consumption per person; $u(c_t)$: utility per person.
- ρ : the rate of time preference; $e^{-\rho t}$ discounts future utility.
- Household size grows at rate n .

$$u(c) = \frac{c^{1-\theta}}{1-\theta}, \quad u'(c) = c^{-\theta} > 0, \quad u''(c) = -\theta c^{-\theta-1} < 0.$$

$u'(c) \rightarrow \infty$ as $c \rightarrow 0$; $u'(c) \rightarrow 0$ as $c \rightarrow \infty$. For $\theta = 1$, use the logarithmic-utility limit.

Aside: expressing utility in effective units

Since $c_t = A_t \hat{c}_t$ and $A_t = A_0 e^{xt}$,

$$u(c_t) = \frac{(A_t \hat{c}_t)^{1-\theta}}{1-\theta} = \frac{\hat{c}_t^{1-\theta}}{1-\theta} A_0^{1-\theta} e^{(1-\theta)xt}.$$

Also $L_t/H = (L_0/H)e^{nt}$. Hence

$$U = \frac{L_0 A_0^{1-\theta}}{H} \int_0^\infty \frac{\hat{c}_t^{1-\theta}}{1-\theta} e^{-[\rho-n-(1-\theta)x]t} dt.$$

The positive constant $L_0 A_0^{1-\theta}/H$ does not affect the optimal choices. We may normalize $L_0/H = 1$ and $A_0 = 1$.

Aside: deriving the budget constraint

In per-person units,

$$c_t + i_t = w_t + r_t k_t, \quad i_t = \dot{k}_t + (\delta + n)k_t.$$

Since $k_t = A_t \hat{k}_t$, differentiation gives

$$\dot{k}_t = A_t \dot{\hat{k}}_t + x A_t \hat{k}_t.$$

Divide by A_t :

$$\hat{i}_t = \dot{\hat{k}}_t + (\delta + n + x)\hat{k}_t.$$

Substitute into $\hat{c}_t + \hat{i}_t = w_t/A_t + r_t \hat{k}_t$:

$$\dot{\hat{k}}_t = w_t/A_t + (r_t - \delta - n - x)\hat{k}_t - \hat{c}_t.$$

This completes the aside.

The household's problem

$$\max_{\{\hat{c}_t, \hat{k}_t\}_{t \geq 0}} \int_0^{\infty} \frac{\hat{c}_t^{1-\theta}}{1-\theta} e^{-[\rho-n-(1-\theta)x]t} dt.$$

Subject to

$$\dot{\hat{k}}_t = w_t/A_t + (r_t - \delta - n - x)\hat{k}_t - \hat{c}_t$$

$$\hat{k}_0 \text{ given}$$

$$\theta > 0, \quad \rho > 0, \quad \rho - n - (1 - \theta)x > 0$$

Optimal Control Theory: The Hamiltonian

Define $\gamma \equiv \rho - n - (1 - \theta)x > 0$. The **present-value Hamiltonian** is

$$\mathcal{H}_t = \frac{\hat{c}_t^{1-\theta}}{1-\theta} e^{-\gamma t} + \psi_t \left[\frac{w_t}{A_t} + (r_t - \delta - n - x)\hat{k}_t - \hat{c}_t \right].$$

- Control variable: $\hat{c}_t$.
- State variable: $\hat{k}_t$.
- Costate variable: ψ_t , the present-value shadow value of wealth.

The household takes the paths of w_t and r_t as given.

The maximum principle

An interior optimum satisfies

$$\frac{\partial \mathcal{H}_t}{\partial \hat{c}_t} = 0,$$

$$\frac{\partial \mathcal{H}_t}{\partial \psi_t} = \dot{\hat{k}}_t.$$

$$-\frac{\partial \mathcal{H}_t}{\partial \hat{k}_t} = \dot{\psi}_t,$$

Together with the initial condition and the transversality condition:

$$\lim_{t \rightarrow \infty} \psi_t \hat{k}_t = 0.$$

The transversality condition rules out leaving positive shadow-valued wealth unused in the limit.

First-order conditions

$$\hat{c}_t^{-\theta} e^{-\gamma t} - \psi_t = 0 \quad \Rightarrow \quad \psi_t = \hat{c}_t^{-\theta} e^{-\gamma t}, \quad (8)$$

$$\dot{\psi}_t = -\psi_t(r_t - \delta - n - x), \quad (9)$$

$$\dot{\hat{k}}_t = w_t/A_t + (r_t - \delta - n - x)\hat{k}_t - \hat{c}_t. \quad (10)$$

The first condition equates the co-state variable to the discounted marginal utility of consumption. Hence it is the shadow value of wealth.

Deriving the Euler equation: differentiate

Differentiate $\psi_t = \hat{c}_t^{-\theta} e^{-\gamma t}$:

$$\begin{aligned}\dot{\psi}_t &= -\theta \hat{c}_t^{-\theta-1} \dot{\hat{c}}_t e^{-\gamma t} - \gamma \hat{c}_t^{-\theta} e^{-\gamma t} \\ &= -\theta \psi_t \frac{\dot{\hat{c}}_t}{\hat{c}_t} - \gamma \psi_t.\end{aligned}$$

Divide by $\psi_t > 0$:

$$\frac{\dot{\psi}_t}{\psi_t} = -\theta \frac{\dot{\hat{c}}_t}{\hat{c}_t} - \gamma. \quad (11)$$

Deriving the Euler equation: substitute

Combine (9) and (11):

$$\begin{aligned}-\theta \frac{\dot{\hat{c}}_t}{\hat{c}_t} - \gamma &= -(r_t - \delta - n - x), \\ \theta \frac{\dot{\hat{c}}_t}{\hat{c}_t} &= r_t - \delta - n - x - [\rho - n - (1 - \theta)x] \\ &= r_t - \delta - \rho - \theta x.\end{aligned}$$

Therefore

$$\boxed{\frac{\dot{\hat{c}}_t}{\hat{c}_t} = \frac{1}{\theta}(r_t - \delta - \rho - \theta x).} \quad (12)$$

In per-person units, $\dot{c}_t/c_t = (r_t - \delta - \rho)/\theta$.

Competitive equilibrium: two dynamic equations

Substitute $r_t = \alpha \hat{k}_t^{\alpha-1}$:

$$\frac{\dot{\hat{c}}_t}{\hat{c}_t} = \frac{1}{\theta} [\alpha \hat{k}_t^{\alpha-1} - \delta - \rho - \theta x]. \quad (13)$$

Use $w_t/A_t = (1 - \alpha) \hat{k}_t^\alpha$ and $r_t \hat{k}_t = \alpha \hat{k}_t^\alpha$:

$$\dot{\hat{k}}_t = \hat{k}_t^\alpha - (\delta + n + x) \hat{k}_t - \hat{c}_t. \quad (14)$$

These are two nonlinear differential equations in $\hat{c}_t$ and $\hat{k}_t$. We study them using a phase diagram and, subsequently, a log-linear approximation around the steady state.

The steady state

For positive consumption, $\dot{\hat{c}}_t = 0$ implies

$$\alpha \hat{k}_{ss}^{\alpha-1} = \delta + \rho + \theta x,$$
$$\boxed{\hat{k}_{ss} = \left(\frac{\alpha}{\delta + \rho + \theta x} \right)^{1/(1-\alpha)}}. \quad (15)$$

Setting $\dot{\hat{k}}_t = 0$ gives

$$\boxed{\hat{c}_{ss} = \hat{k}_{ss}^{\alpha} - (\delta + n + x) \hat{k}_{ss}}. \quad (16)$$

The effective discount condition implies $\rho + \theta x > n + x$, ensuring an interior steady state with positive consumption.

Phase diagram: the two loci

$\dot{\hat{c}} = 0$: **a vertical line**

$$\hat{k} = \hat{k}_{ss}.$$

- $\hat{k} < \hat{k}_{ss}$: $\dot{\hat{c}} > 0$.
- $\hat{k} > \hat{k}_{ss}$: $\dot{\hat{c}} < 0$.

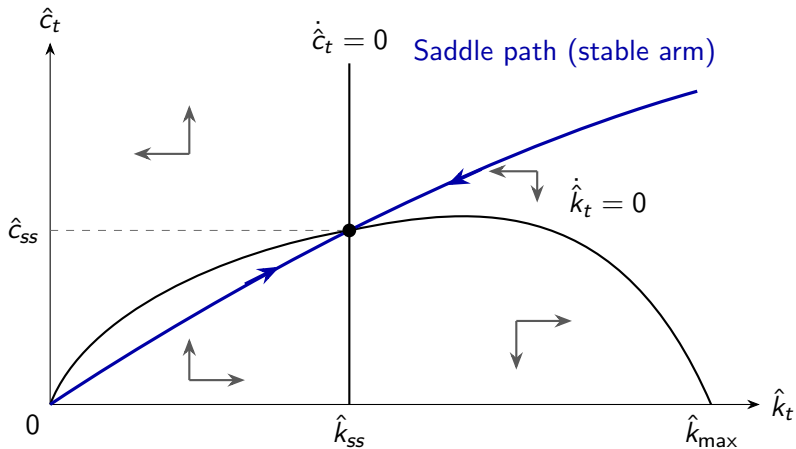
$\dot{\hat{k}} = 0$: **a concave curve**

$$\hat{c} = \hat{k}^\alpha - (\delta + n + x)\hat{k}.$$

- Below the curve: $\dot{\hat{k}} > 0$.
- Above the curve: $\dot{\hat{k}} < 0$.

The loci intersect at $(\hat{k}_{ss}, \hat{c}_{ss})$. The positive horizontal intercept is $\hat{k}_{\max} = [1/(\delta + n + x)]^{1/(1-\alpha)}$.

Phase diagram and saddle path



Why not choose a path below the saddle path?

A path below the saddle path involves excessive saving.

- It approaches $\hat{k}_{\max}$ with $\hat{c} \rightarrow 0$.
- The saving rate approaches one; marginal utility becomes unbounded.
- Consumption could be reallocated across time to improve welfare.

At the limiting capital stock,

$$\alpha \hat{k}_{\max}^{\alpha-1} = \alpha(\delta + n + x),$$

so the costate growth rate approaches

$$\frac{\dot{\psi}}{\psi} = (1 - \alpha)(\delta + n + x) > 0.$$

Thus $\psi_t \hat{k}_t$ fails to approach zero: the transversality condition is violated.

Paths above the saddle path and the optimal path

Above the saddle path:

- The trajectory eventually enters a region with $\dot{\hat{c}} > 0$ and $\dot{\hat{k}} < 0$.
- Rising consumption and a declining capital stock eventually exhaust available resources.
- The path cannot remain feasible indefinitely.

On the saddle path:

- $(\hat{k}_t, \hat{c}_t) \rightarrow (\hat{k}_{ss}, \hat{c}_{ss})$ asymptotically.
- The Euler equation and resource constraint hold throughout.
- The transversality condition also holds, as shown next.

The Golden Rule level of capital

Choose capital to maximize steady-state consumption:

$$\max_{\hat{k}} \hat{c} = \hat{k}^{\alpha} - (\delta + n + x)\hat{k}.$$

The first-order condition is

$$\alpha \hat{k}^{\alpha-1} = \delta + n + x.$$

Hence

$$\hat{k}_{gr} = \left(\frac{\alpha}{\delta + n + x} \right)^{1/(1-\alpha)}.$$

Since $\rho - n - (1 - \theta)x > 0$,

$$\delta + \rho + \theta x > \delta + n + x \quad \Rightarrow \quad \hat{k}_{ss} < \hat{k}_{gr}.$$

Golden Rule and the transversality condition

The costate equation implies

$$\frac{\dot{\psi}_t}{\psi_t} = -[\alpha \hat{k}_t^{\alpha-1} - \delta - n - x].$$

Therefore

Capital stock	$\dot{\psi}_t/\psi_t$
$\hat{k}_t < \hat{k}_{gr}$	< 0
$\hat{k}_t = \hat{k}_{gr}$	$= 0$
$\hat{k}_t > \hat{k}_{gr}$	> 0

At the Ramsey steady state,

$$\frac{\dot{\psi}_t}{\psi_t} \rightarrow -[\rho - n - (1 - \theta)x] = -\gamma < 0.$$

Since $\hat{k}_t \rightarrow \hat{k}_{ss}$ and $\psi_t \rightarrow 0$, $\psi_t \hat{k}_t \rightarrow 0$.

Long-run growth properties

On the balanced growth path, effective-worker variables are constant.

Variable	Long-run growth rate
$\hat{k}, \hat{c}, \hat{y}, \hat{i}$	0
$K/L, C/L, Y/L, I/L$	x
K, C, Y, I	$n + x$

The net return to capital is constant:

$$r_{ss} - \delta = \alpha \hat{k}_{ss}^{\alpha-1} - \delta = \rho + \theta x.$$

Capital deepening and technological progress have offsetting effects on the return to capital.

Income levels, growth, and convergence

On the balanced growth path,

$$\frac{Y_t}{L_t} = A_t \hat{k}_{ss}^\alpha = A_0 e^{xt} \left(\frac{\alpha}{\delta + \rho + \theta x} \right)^{\alpha/(1-\alpha)}.$$

- Similar long-run income levels require similar technology and structural parameters, including ρ and θ .
- Sustained high or low growth is driven by high or low x .
- Holding the other parameters fixed, a tenfold difference in A_t produces a tenfold difference in income per person.
- Preferences affect saving and capital accumulation, and therefore income levels.

The endogenous saving rate

Define the gross saving/investment rate:

$$s_t = \frac{\hat{i}_t}{\hat{y}_t} = 1 - \frac{\hat{c}_t}{\hat{y}_t}.$$

In the steady state, $\hat{i}_{ss} = (\delta + n + x)\hat{k}_{ss}$, so

$$\begin{aligned} s_{ss} &= \frac{(\delta + n + x)\hat{k}_{ss}}{\hat{k}_{ss}^\alpha} \\ &= (\delta + n + x)\hat{k}_{ss}^{1-\alpha} \\ &= \boxed{\frac{\alpha(\delta + n + x)}{\delta + \rho + \theta x}}. \end{aligned}$$

The Ramsey model determines the saving rate from preferences, technology, and population growth.

Saving: preference and population effects

$$s_{ss} = \frac{\alpha(\delta + n + x)}{\delta + \rho + \theta x}.$$

- $\rho \uparrow$: households discount future utility more heavily; $s_{ss} \downarrow$.
- $\theta \uparrow$: utility is more curved and the intertemporal elasticity of substitution $1/\theta$ falls. For $x > 0$, $s_{ss} \downarrow$.
- $n \uparrow$: more investment is needed to equip a growing population; $s_{ss} \uparrow$.

Because saving adjusts, $\hat{k}_{ss}$ and $\hat{y}_{ss}$ do not depend on n in this specification. If $x = 0$, θ does not affect s_{ss} .

Saving during the transition

The constant saving rate is a long-run result; it need not hold along the development path. During development, s_t may rise or fall toward s_{ss} .

- **Substitution effect:** as $\hat{k}$ rises, $r - \delta$ falls. Saving becomes less attractive, tending to lower s_t .
- **Income effect:** households smooth consumption as income rises. Output may initially grow faster than consumption, lowering $\hat{c}/\hat{y}$ and raising s_t .
- The two effects work in opposite directions.
- θ influences the strength of consumption smoothing and thus the transition of the saving rate.